

Demand responsive transport: New insights from *peri*-urban experiences

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ABSTRACT

Despite great differentials between urban, *peri*-urban and rural experiences, little research on demand responsive transport (DRT) has been conducted in *peri*-urban settings. This paper explores the influence of *peri*-urban area (PUA) conditions on the design and the operation of DRT and the resulting demand patterns and modal shifts. It uses a combination of a trip dataset and a survey of users and non-users. Key findings suggest that the provision of public transport services within the PUA and to nearby towns and cities, the high car ownership rate and the low demand have a significant influence in designing the DRT scheme. An integrated DRT and public transport (PT) system has potentials to be successful in a context of PUA. Our empirical evidence indicates that such a DRT service design can improve the accessibility and generate a stable demand while reducing the risk of substituting PT. However, there is a danger of the unwanted substitution of walking and cycling. We provide recommendations for DRT planners and operators in developing future schemes in PUA settings.

1. Introduction

The *peri*-urban area (PUA), once considered as a forgotten and bypass place (Sieverts, 2012) in the dichotomy of the rural–urban divide, has been increasingly seen as a distinctive space in its own right (Rauws and De Roo, 2011). It is no longer only viewed as a space in-between of the city and its countryside, but also a dynamic and multifunctional territory that provides a terrain for new experiments and innovations (Sieverts, 2012). Amid of this transitional characteristic, mobility and infrastructures have always been a key component in shaping the nature of the PUA. Yet the historical heritages coupled with the globalization development usually make many neighborhoods in the PUA disconnected by public transport (PT) and thus car dependency (Young and Keil, 2010).

It is not uncommon that in certain *peri*-urban neighborhoods PT services are poor (Young and Keil, 2010) or non-existent even in such an affluent country as Switzerland. Consequently, PT dependent user groups, including those on low incomes, the young and the elderly people face the problem of first-/last-mile connections (Thao et al., 2021). Such barriers create a great challenge to achieving social equality and sustainable mobility (Docherty et al., 2018). Meanwhile, public-sector bodies and PT operators have traditionally been criticized for being less agile in meeting the demands of diverse user groups (Sweet, 2021). Collaboration among the public, private and voluntary sectors in mobilizing integrated resources may overcome these barriers (Brinkerhoff and Brinkerhoff, 2011).

Transport systems are undergoing a transition toward smart mobility

in which mobility is characterized as a personalized service and is made available ‘on demand’, thanks to advanced technologies (Docherty et al., 2018), new attitudes toward mobility-on-demand services and demographic changes (Shaheen and Cohen, 2020). This paradigm shift appears to offer an innovative mobility option for PT problems in low demand areas (Kameda and Mukai, 2011). Although there is great potential and a need for mobility on demand not only to fill gaps in equity and exclusion in relation to mobility, but also to contribute to environmental sustainability in low demand areas (Fournier et al., 2018; Gross-Fengels and Fromhold-Eisebith, 2018), the current literature on mobility on demand often focuses on ride-hailing, ride-sharing and shared automated vehicles in cities (for examples, see Cohen-Blankshain and Rotem-Mindali, 2016; Acheampong et al., 2020) or on demand responsive transport (DRT) in rural areas (Gross-Fengels and Fromhold-Eisebith, 2018; Kameda and Mukai, 2011; Nyga et al., 2020). To date, few studies (see Kaufman et al., 2021; Teal and Becker, 2011) on DRT has been conducted in *peri*-urban settings. Some studies (e.g., Mageean and Nelson, 2003) covered these regions but did not treat them as a separate geographical context and granted them little attention in the analysis. Furthermore, previous studies of DRT tend to focus on the supply side (Davison et al., 2012) and less on the demand side (Davison et al., 2012; Haglund et al., 2019). There is also a lack of knowledge about non-user groups, although they can provide a broader understanding of the market demand for DRT services (Wang et al., 2014).

Despite its diversity, the PUA shares some commonalities that may influence the effectiveness in implementation of DRT services. Furthermore, transportation interventions like DRT implementations

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can benefit from experiences taken elsewhere, given the failure of many promising DRT schemes (Davison et al., 2012). Against this backdrop, the current paper seeks to answer the following research questions: (1) How are the design and the operation of a DRT scheme in a PUA influenced by a specific set of *peri-urban* area conditions? (2) What impacts do they have on demand patterns and modal shifts? The analysis draws upon the EBuxi pilot project in two *peri-urban* municipalities in Switzerland, using a combination of two datasets, a survey of (non)users and trip data from the operator. It is noteworthy that using the operator's data for investigating demand patterns of DRT trips has been done in a few studies due to lack of access (Haglund et al., 2019; Sanaullah et al., 2021). Having this dataset could provide insights on the impact of particular *peri-urban* area conditions on the usage behavior of DRT services, instead of only relying on user survey data as in most of the studies.

Our exploratory study can help to inform policy-makers and public transit agencies in designing and implementing DRT services in the PUA. It does so by documenting the way in which municipalities, private companies, civil-society associations and voluntary drivers mobilize their financial and human resources in developing and operating DRT services, thus promoting the viability of such transport solutions.

The remainder of the paper is organized as follows. Section 2 provides characteristics of PUAs and factors influencing the demand pattern of DRT services. Section 3 presents the case selection and introduces the datasets and methods used. Section 4 describes the two *peri-urban* neighborhoods studied and how EBuxi as a flexible and shared DRT was founded, designed and operated by a collaboration of public, private and voluntary stakeholders. Section 5 presents results, while section 6 discusses the key findings and provides implications for management by PT authorities, policy-makers and DRT operators.

2. Literature review

2.1. Peri-urban characteristics

Despite a growing scholarly interest, a consensus on the definition of the PUA is still missing. What agreed in the literature is that these regions share some commonalities (Shaw et al., 2020). The PUA has a mix of rural and urban features, thus often being viewed as an in-between space. It has a relatively low population density compared with the urban's standards. There is a strong interdependence between the PUA and the city nearby. The PUA has multiple functionals for the city such as the supply of labor, cheap housing, and spaces for industrial manufactures (Sieverts, 2012). Concurrently, the PUA is dependent on the facilities and services of the city. The interdependence and the geographical proximity result in a high commuting demand (Ravetz et al., 2013). However, the provision of transport infrastructure is a major problem in PUAs. Inhabitants in many in-between spaces strongly rely on cars to meet their mobility needs due to lack of PT services. Those who do not have access to cars face risks of isolation and social exclusion (Young and Keil, 2010). Several authors therefore called for the provision of alternative infrastructure connectivity but acknowledged the governance challenges because of its blurred spatial boundary (Sieverts, 2012; Young and Keil, 2010). To safeguard wellbeing in the PUA, planning policies and strategies in the provision of transport infrastructure require flexibility (Sieverts, 2012) and must anticipate its dynamics.

2.2. Factors influencing the demand for and usage behavior of DRT services

Mageean and Nelson (2003) defined DRT as “an intermediate form of transport, somewhere between a regular bus and a taxi” (p. 255). With the appearance of mobile technology and real-time information, DRT services are increasingly applying flexible routing systems and non-predefined schedules (Jain et al., 2017; Sanaullah et al., 2021). DRT

services often serve less populated areas where the provision of conventional bus services is not cost-effective because of low demand (Wang et al., 2015). The services can be designed to expand or replace bus services or as a new transport service. It can also be used as a solution to the first-/last-mile problem. With flexible routes and the flexible scheduling of pick-up timings, they can increase service coverage, ridership and cost-efficiency (Brake et al., 2004), as well as improving social inclusion and community building (Nelson and Phonphitakchai, 2012).

Wang et al., (2014) reported that the demand for DRT services in the city region of Greater Manchester in the UK was higher in areas with lower population density and car ownership. Similarly, Nelson and Phonphitakchai (2012) found that the most successful DRT scheme (LinkUp) operated in an urban context of England where the car ownership rate was very low. In rural contexts where car ownership is general high the DRT services may encounter a slow initial growth of the demand as empirical evidence in a rural community West Sussex in the UK shows (Brake et al., 2004). The dispersal of activities centers also affect the usage pattern. Examining a flexible on-demand micro-transit service in the Helsinki capital region of Finland, Haglund et al. (2019) observed a strong dominance of trips between the periphery and the city center as well as several business locations during the morning and evening peak hours. A study of three different DRT schemes in lower density suburban zones of Denver in the US shows that transport infrastructures (e.g., having a local railway station and poor street connectivity), the dispersal of points of interest such as shopping centers and large office buildings, and residential areas with young and relatively low income families impact the ridership and trip purposes of the DRT services (Teal and Becker, 2011). Another DRT study conducted in a *peri-urban* context in Australia found no correlation between the population density and the trip numbers across three suburbs but the specific conditions of PT services play a crucial role (Kaufman et al., 2021). In short, the above discussion suggests that transport infrastructure and quality, the diversity of land-use and employment concentration may determine demand patterns for DRT services in a *peri-urban* context.

Service design features can influence the demand for DRT services too. They can include the ease of booking methods, the degree of flexibility of routes and timetables (Brake et al., 2004; Mageean and Nelson, 2003), operating times (Nelson and Phonphitakchai, 2012), the fleet size and the coverage of operating areas (Haglund et al., 2019). The individual level factors such as age, gender, having car access (Jain et al., 2017; Nelson and Phonphitakchai, 2012), values, awareness and attitudes towards the services (Nelson and Phonphitakchai, 2012) may also have an impact on the DRT demand.

Research on the impact of DRT services on modal shifts is limited so far. Previous studies on this topic mainly used a simulation approach. Empirical evidence in urban contexts of Netherlands indicates that DRT can complement public transit, given that the design of DRT services as an integration in PT systems (Alonso-González et al., 2018).

3. Case selection and methodology

3.1. Case selection

In recent years, several DRT schemes have been introduced in Switzerland. They have different service designs depending on the institutional and socio-spatial contexts. EBuxi was initiated and operated by a private actor and a voluntary organization, while all other DRT schemes founded and operated by PT operators. The service was first introduced in Herzogenbuchsee and Niederönz *peri-urban* municipalities, and later expanded in other regions. For the research purposes of this study, we selected the EBuxi DRT in the municipalities.

3.2. Data collection

Information on the influence of PUA qualities on the design and the operation of EBuxi scheme was obtained through three semi-structured interviews with the mayor of Herzogenbuchsee municipality, a member of EBuxi association's steering committee and the founder of mybuxi start-up company. The interviews were conducted in March 2020 and lasted more than one hour.

For purposes of the analysis of demand patterns and modal shifts, two datasets were used. The first dataset was obtained from the EBuxi company. It consists of 31,779 anonymized trips recorded from 28 April 2019 to 27 July 2021 (no trips were made during the coronavirus lockdown from 18 March 2020 to 27 April 2020). Of these, 2,111 trips were canceled or were no shows, which we excluded from our analysis. We used 29,668 successful trips. Each of these trips contains the following information: date of the trip; pick-up and drop-off time; geo-coordinates of pick-up and drop-off places; and route path. Based on the geocoordinates of pick-up and drop-off places, the origin and destination of the trip were identified using Quantum Geographic Information System. The pick-up and drop-off times were used to determine the temporal demand of daily trips that could be useful for planning the operation of DRT services. The geographical distribution of trips was used to identify a relationship between *peri*-urban attributes and user demand patterns.

The second dataset is a survey of users and non-users of EBuxi services that we designed and administered. The questionnaire survey contains two parts. The first part covers common questions on access to/ownership of mobility tools (e.g., car, PT season ticket) and elicits basic demographics. The second part consists of specific questions for users and non-users of the service. Respondents were filtered by their answer to the question "Have you ever used the EBuxi service?" Users and non-users were then asked to choose their reasons for using and not using EBuxi from a list of options administered separately to each group. For the users, the survey asked questions regarding the reasons for using the EBuxi service, trip purposes and frequencies, and previous, following and alternative modes of transport. For the non-users, the survey asked questions regarding the reasons for not using the service. This data can be helpful for what Wang et al.'s (2015) claimed about a little knowledge on the potential market niches for DRT.

Following a pretest, the survey was conducted in spring 2020. To reach the largest number of potential respondents, we deployed both online and offline distribution channels targeting respondents above 16 years old. An invitation to participate in our survey was sent to all subscribers of the EBuxi association's newsletter. The EBuxi association also informed their drivers about the survey and instructed them to hang flyers in EBuxi minibuses. EBuxi users could either take the flyer with them or use the QR code in the flyers to answer our survey. In addition, we sent letters by post with a simple weblink to the survey to 2,000 (49% of total population) and 400 households (53% of total population) in Herzogenbuchsee and Niederönz respectively. In total, 520 questionnaires were collected, achieving a 21.7% response rate. Of these 470 were completed and usable, and 50 were excluded from the analysis because of major failures to complete. As our survey was conducted online, people without access to internet might be underrepresented in our sample, thus leading to biased results. However, when compared with the whole Swiss population (FSO, 2021a, 2021b), our sample presented in Table 2 is representative.

3.3. Data analysis

Table 1 summarizes how the methods used to collect data and their corresponding analytical techniques contribute to research questions. The trip dataset was used to analyze the development of the demand and the geographical and daily travel time distributions of trips. The survey dataset was used to understand usage behavior, namely the characteristics of users and non-users, the reasons for using and not using the

Table 1

Methods, data collection and analytical techniques.

Research questions	Data	Analytical techniques
Influence of <i>peri</i> -urban conditions on the design and the operation of the DRT service	Interview	Theme analysis
Resulting demand patterns		
- User and non-user profile	- Survey	- Descriptive analysis
- Motivation for use and not use	- Survey	- Descriptive analysis
- Factors influence adoption of the service	- Survey	- Logistic regression modeling
- Demand and usage behavior patterns	Survey, Trip dataset	- Descriptive analysis
Potential impacts on modal shifts	Survey	Descriptive analysis

service, trip purposes, and the willingness to wait or to detour. This dataset was also used to explore the potential impacts of the EBuxi service on modal shifts.

For the statistical analyses, the SPSS 28.0 software was used. To identify factors influencing the adoption of the EBuxi service, logistic regression modeling was adopted using the survey data. The dependent variable of the adoption of the EBuxi service is coded binary with a discrete outcome (yes or no). Mobility tools and various sociodemographic characteristics are used as predictors. Table 2 presents the operationalization of data, measurement, and sample characteristics for the logistic regression modeling.

A multicollinearity test was conducted. No multicollinearity of independent variables in the model was found, as all the pairwise correlations were below a threshold of 0.7. The overall significance of the model was tested with a χ^2 test. To assess goodness of fit, the Hosmer-Lemeshow test was conducted. Nagelkerke's and Cox & Snell's R^2 were used to represent the proportion of the variance for the depended variable that is explained by the independent variables in the regression model.

4. Case description

Herzogenbuchsee and Niederönz municipalities are situated in the north-eastern part of the canton of Berne, Switzerland. They are characterized as *peri*-urban areas (FSO, 2017). In 2019, Herzogenbuchsee had a population of 7,255 and a population density of 739 people/km². With 1,681 inhabitants, the neighboring municipality of Niederönz is much smaller, and it also has a lower population density (603 people/km²). Herzogenbuchsee has more incoming labor than outcoming labor, while the opposite is true for Niederönz.

The municipalities have typical qualities of PUAs. The municipalities have good railway connections to nearby towns and cities (see Fig. 1). Local people can travel to Berne city in 30 min from the regional railway station. This railway connection runs every half hour. Another connection takes less than 20 min to reach one of the country's mobility hubs, Olten railway station. However, the PT service within the municipalities is rather poor. There are four bus lines, two of which operate every quarter of an hour on weekdays, while the other two operate hourly. At weekends, one bus line runs every half an hour and each of the other three bus lines operates either hourly or every-two or three hours. Due to the low population densities, large residential areas of both Herzogenbuchsee and Niederönz are not served by a bus service. This can contribute to the high car ownership rate in the municipalities.

Herzogenbuchsee has several urban center functions that also serve Niederönz's population. There are three small industrial factories whose workers come from nearby regions. A regional employment center and a retirement resident are also located in the municipality. Other points of interest like a clinic for addiction treatment, sport center, a large public swimming pool attract users from surrounding municipalities. They are scattered and far from the village center where the regional railway

Table 2
Summary of variables for modeling.

Dependent variable			
Adoption	The service was used once or more	Yes	0 = No – 42.3% 1 = Yes – 57.7%
Independent variables			
	Data description	Focus	Measurement Level and Type, Statistics
Sociodemographic characteristics			
Gender	Men/Women	Man	Nominal: 0 = Woman – 53% 1 = Man – 47%
Age	Age group: 16–34; 36–64; 65+	16–34 35–64	Ordinal: 16–34: 21.1% 35–64: 53% 65+: 26%
Education level	Higher education (university, college)Other (compulsory school, upper secondary school, vocational degree)	Higher education	Nominal: 0 = Others – 57.9% 1 = Higher Education – 42.1%
Occupation	EmployedOther (unemployed, retired, housekeeping, in education)	Employed	Nominal: 0 = Others – 37.9% 1 = Employed – 62.1%
Mobility tools			
Car availability	Frequent or constant access to car (Yes/No)	Yes	Nominal: 0 = No – 25.5% 1 = Yes – 74.5%
Bicycle availability	Constant access to bicycle (Yes/No)	Yes	0 = No – 69.8% 1 = Yes – 30.2%
PT season tickets	General-, year-, month-, and week-ticket (Yes/No)	Yes	0 = No – 38.9% 1 = Yes – 61.1%

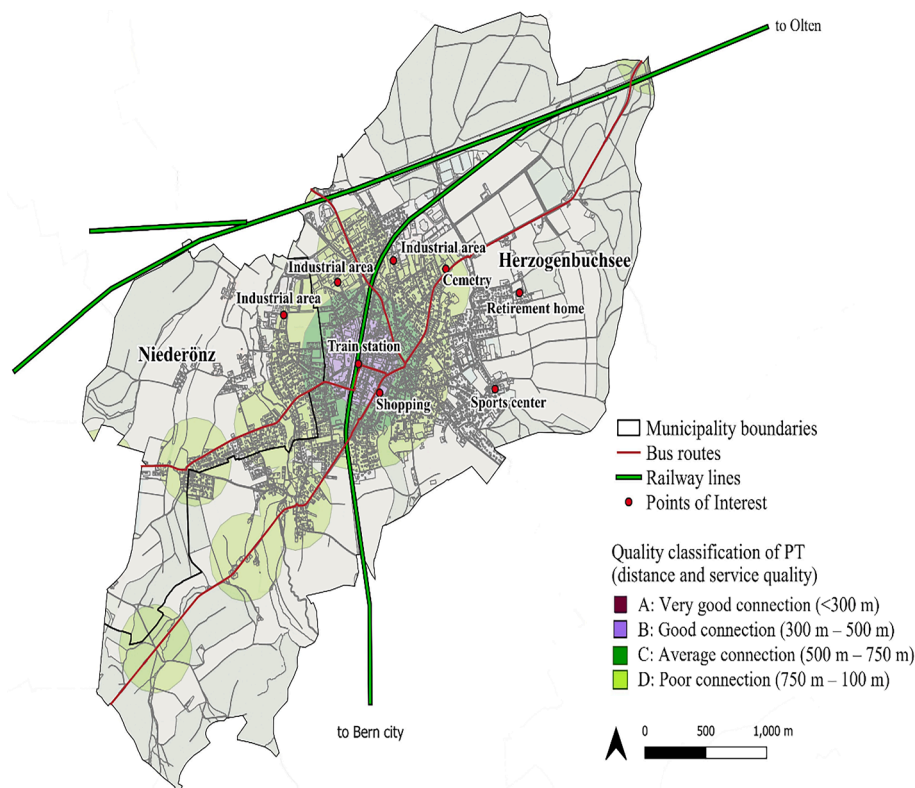


Fig. 1. Accessibility quality in the research area.

station is located. The distance from the station to these points is usually more than one kilometer but walking was the common option because PT connections and a local taxi service are not existed. To improve accessibility and promote intermodal trips, a DRT scheme (EBuxi) was developed as a first-/last-mile solution. Hence, even though the core

concept of EBuxi is to provide a door-to-door, shared, point-to-point transportation on demand with flexible routes and flexible schedules, it has a predefined stop and its own timetable at the railway station (See Plate 1). At this stop, customers can order a ride spontaneously without having to make a booking in advance, either via a smartphone



Plate 1. EBuxi bus charging at the railway station, next to its timetable.

application or by telephone. The EBuxi service was designed to run from the first until the last train service and open to anybody to use. EBuxi also optimizes its pooled rides for customers who have bus connections right after the EBuxi ride. With the above specific *peri*-urban conditions, EBuxi was developed as a feeder service to trains and buses.

Being categorized as a *peri*-urban municipality, Herzogenbuchsee received limited financial supports from the government for transport development. Together with Niederörsz municipality and the mybuxi start-up, a private company looking for a pilot region in which to test its IT services for flexible and shared DRT, Herzogenbuchsee applied for a public grant. The project was granted for one year. The influence of the local conditions is also reflected in scheme design. Minibuses of seven seats were chosen because this type provides cost effective as a low demand for the service was anticipated due to the high car ownership rate. Moreover, operating this vehicle type does not require professional bus driver licenses, hence opening an opportunity for engaging volunteer drivers which in turn reduces operating costs.

The EBuxi association was established in 2018 to support the operation of the EBuxi service in Herzogenbuchsee and Niederörsz. The association is responsible for the acquisition and coordination of around 40 volunteer drivers and the main financing of the operations. The volunteer drivers receive payments that are far below normal pay rates. They work shifts lasting 4 to 5 h and usually take over two to three shifts per month. The volunteer drivers also take care of bookings via the telephone. A majority of them are retired persons.

A fare per trip costs 3 Swiss Francs in the first service year and 4 CHF from the second year. Note that trips on Saturdays during August 2019 and July 2021 were available free of charge thanks to a local insurance company sponsoring them. Due to current regulations, passengers are not allowed to use their existing PT card for EBuxi service. The EBuxi pilot project started running at the end of April 2019. Thanks to the success of pilot year, the project has started its roll-out phase. EBuxi has received a subsidy in a form of cash contribution from the municipalities for both the pilot and the roll-out phases. For the pilot phase, it also received a financial support by a local bank. A local energy provider sponsored the two electric minibuses and a charging station at the railway station (see Plate 1).

5. Findings

5.1. Characteristics of users and non-users of EBuxi

The survey included 470 respondents, of whom more than half (57.7%) were users and 42.3% were non-users of the EBuxi service. Table 3 shows the profile of users and non-users. The most significant difference between the two groups is the access to mobility tools. More than one-third of the user respondents had no frequent or constant access to cars and bicycles, and nearly-two-thirds of them did not own PT season tickets. For the non-user group, the figures are 18.1%, 23.1%, and 50.3%, respectively. More retired and elderly people were found in

Table 3

Characteristics of users and non-users.

	User (%) (n = 271)	Non-user (%) (n = 199)	Total (%) (n = 470)
Gender			
Female	52.4	53.8	53.0
Male	47.6	46.2	47.0
Age			
16 – 34	18.5	24.6	21.1
35 – 64	48.7	58.8	53.0
65+	32.8	16.6	26.0
Education level			
Compulsory school	4.8	7.0	5.8
Upper secondary school	10.0	11.1	10.4
Vocational degree	39.9	44.2	41.7
Higher education	45.4	37.7	42.1
Occupation			
Employed	57.9	67.8	62.1
Unemployed	1.8	1.0	1.5
Retired	29.2	15.1	23.2
Housekeeping	3.0	6.0	4.3
In education	8.1	10.1	8.9
Car availability			
Yes	69.0	81.9	74.5
No	31.0	18.1	25.5
Bicycle availability			
Yes	64.6	76.9	69.8
No	35.4	23.1	30.2
PT season tickets			
Yes	30.6	49.7	61.1
No	69.4	50.3	38.9

the user group. This suggests that the DRT service was used by people who were disadvantaged in some aspect.

5.2. Reasons for using and not using EBuxi

As seen in Table 4, the top three reasons for using EBuxi are related to supply-side factors, namely reasonable price (84.8%), shorter travel times (63.3%) and high levels of comfort (48.9%). Even though the fare per trip is double that of a short bus ride, it is still much cheaper than a taxi ride. Furthermore, travelling with EBuxi on Saturdays was free of charge for two years.

Based on the trip dataset, the average distance and travel time of EBuxi trips are around 1.25 km and 4.5 min respectively. The distance is certainly quite long for walking. Except for ease of use, the following reasons for using EBuxi are associated with demand-side factors. Respondents also used the EBuxi service because they had no access to a car (39.3%) or to PT (33.7%). Though not so common, ease of use (17%), opportunities for social contacts (12%) and substituting for parent taxis (11%) were reported by the respondents as reasons for using EBuxi.

Almost all the reasons for not using EBuxi are related to demand-side

Table 4

Reasons for using and not using EBuxi.

Reasons for using (multiple responses possible)	Percent	Reasons for not using (multiple responses possible)	Percent
Reasonable price	84.8	Prefer to cycle	43.7
Shorter travel time than walking	63.3	Prefer to walk	42.7
High comfort	48.9	Prefer to use my private car	32.2
No access to car	39.3	Live near train station/in the center	19.6
No access to PT	33.7	Ordering process too complicated	3.0
Ease of use	17.0	Takes too long for EBuxi to reach my place	2.5
Opportunities for social contact	12.0	Other	21.6
Substitute for parent taxi	11.0		
Other	4.8		

factors. Interestingly, preferring to cycle (43.7%) or to walk (42.7%) are the top reasons for not using the service, followed by preferring to use one's private car (32.2%). Almost 20% of the non-users reported that living near the train station or in the municipal center, and thus having better access to PT services, are reasons for not using EBuxi. A preference for the bus and privacy are among other reasons for not using the service.

5.3. Factors influencing the adoption of EBuxi

The logistic regression model for prediction of the adoption of the EBuxi service is statistically significant (χ^2 (8, $N = 470$) = 52.410, $p < .001$) (see Table 5). Therefore, an analysis can be conducted. Nagelkerke's R^2 suggests that the model can explain 14.2% of the variation in the adoption of EBuxi. The Hosmer-Lemeshow test of goodness of fit indicates that the model is a good fit to the data, as the p value ($p = .31$) is greater than 0.05.

The results show that the effect of age is significant and negative, indicating that younger age groups are less likely to use the EBuxi service. The probability that a person in the age groups of 16 to 34 years ($B = 0.333$, $p = .001$) and 35 to 64 years ($B = 0.438$, $p = .009$) uses the service is reduced by 66.7% and 56.2% respectively, compared to the oldest age group (65 +). Education level ($B = 1.522$, $p = .047$) has a significantly positive influence on the adoption of the EBuxi service. The likelihood that a person with higher education uses EBuxi increases by 52.2% in comparison to those who do not possess higher education. PT season tickets ($B = 2.36$, $p = .001$) reveal a significantly positive impact on the use of EBuxi, while car availability ($B = 0.514$, $p = .009$) has the opposite impact. If a person owns a PT ticket, the probability of using the service is increased by 136%. Conversely, having constant access to a car reduces the probability of using the service by 48.6%. Cohen's f^2 is 0.17, which according to Cohen (1992) corresponds to a medium-size effect. Gender, occupation, and bicycle availability were found to have no significant effect on the adoption of the EBuxi service.

5.4. Demand for EBuxi

Fig. 2 illustrates the demand for the EBuxi service over two and a half years. The demand grew rapidly after EBuxi's launch and reached a peak after seven months. Due to lockdown measures against the spread of

Table 5
Logistic regression predicting likelihood of using EBuxi.

	Coeff.	S.E.	Wald	df	Sig.	Exp (B)
Constant	1.14	0.33	11.61 ***	1	0.001	3.123
Gender (ref.: Female)						
Male	−0.14	0.20	0.446	1	0.504	0.873
Age (ref.: Age 65 +)						
16 – 34	−1.099	0.33	11.18 ***	1	0.001	0.333
35 – 64	−0.83	0.32	6.83 **	1	0.009	0.438
Education level (ref.: Other)						
Higher education	0.42	0.21	3.95 *	1	0.047	1.522
Occupation (ref.: Other)						
Employed	−0.048	0.26	0.034	1	0.854	0.953
Car availability (ref.: No)						
Yes	−0.66	0.33	6.92 **	1	0.009	0.514
Bicycle availability (ref.: No)						
Yes	−0.33	0.24	1.95	1	0.163	0.719
PT season tickets (ref.: No)						
Yes	0.86	0.21	17.579 ***	1	0.001	2.36

χ^2 (8, $N = 470$) = 52.410, $p < .001$.

Nagelkerke $R^2 = 0.142$; Cox & Snell $R^2 = 0.106$; Correctly Classified Cases: 63.8%; Hosmer-Lemeshow $p = .31$

coronavirus infections, the service was suspended for some weeks. The demand increased after reopening the service but did not reach the level it did before the lockdown. An increase in remote working may explain for the lower patronage. On average, EBuxi transported 1,590 rides with 1,880 passengers per month.

Providing services on Saturdays free of charge tends to have little impact on demand. It stimulated the demand after the launch, but ceasing the promotion shows no impact on the demand. The introduction of a free-of-charge service led instead to more 'no show' bookings and very short trips being booked by adolescents. This undesired behavior reduced regular users' satisfaction rates as well as voluntary drivers' motivations (personal communication with the CEO of mybuxi company).

Fig. 3 shows that demand grows sharply in the morning and drops off at lunchtime. It then increases again, with most trips occurring between 13:00 and 18:00. The demand falls slowly again between 18:00 and 24:00. This finding supports the finding on trip purposes, namely that a sustainable number of trips are made to return home in the evenings.

5.5. Geographical distributions of trips and trip purposes

Based on the geocoordinates of the pick-up and drop-off places of each trip, we illustrate the association between *peri*-urban attributes, i. e., the points of interests and trips' geographical distributions (see Fig. 4). For better visualization, the proportions of trip distributions below two percent are not displayed. Overall, the origin–destination flows show that trips were mainly generated in and distributed across the railway station and residential areas. For example, 39.5% of the trips originated at the railway station, and 18.6% ended there. About 22.7% of the trips started and ended in the residential areas. Around 27.6% of the trips started at the railway station and ended in the residential areas. Besides, the origin–destination flows indicate that relatively substantial proportions of trips went across the railway station and the industrial areas, as well as the railway station and the retirement home. These suggest that the EBuxi service was used as a solution to the first-/last-mile problem. It is also worth mentioning that 4.6% and 6.4% of the trips started and ended at the retirement home respectively. This finding strengthens the earlier finding that a significant number of EBuxi users are aged over 65. EBuxi can meet the travel demands of the older user group.

The results of the survey supported the findings mentioned above. When we asked respondents about the purposes of their trips, getting to or from the local railway station was cited by the largest number of respondents (83%). About 48.5% were returning home in evenings. Smaller percentages of respondents reported they used EBuxi for doctor's appointments (18.5%), shopping (11.9%) and visits to retirement homes (7.7%). Other trips (6.3%) were taken to or from industrial areas, the sports center and the regional employment center.

5.6. Willingness to wait and willingness to detour

Fig. 5 presents willingness to wait and willingness to detour. User respondents were asked how long they would be willing to wait from the moment they finished booking an EBuxi trip until being picked up. More than two-thirds would wait up to 15 min. The percentage of the user respondents fell dramatically (18.5%) when the waiting time was increase to between 16 and 30 min. Interestingly, the main user group (35–64 years) was more willing to wait longer, while the youngest group was the least willing to wait. Only 4% said they would be ready to wait for more than 30 min. Note that, based on the trip dataset, the average waiting time is seven minutes.

Rides with similar times and routes are pooled. Therefore, trips may be longer due to detours. We asked the user respondents how long a detour they would tolerate to pick up additional passengers. About 43.5% would accept a detour time of up to 5 min. Nearly half of them would tolerate a detour time of six to ten minutes. Those 65 and older

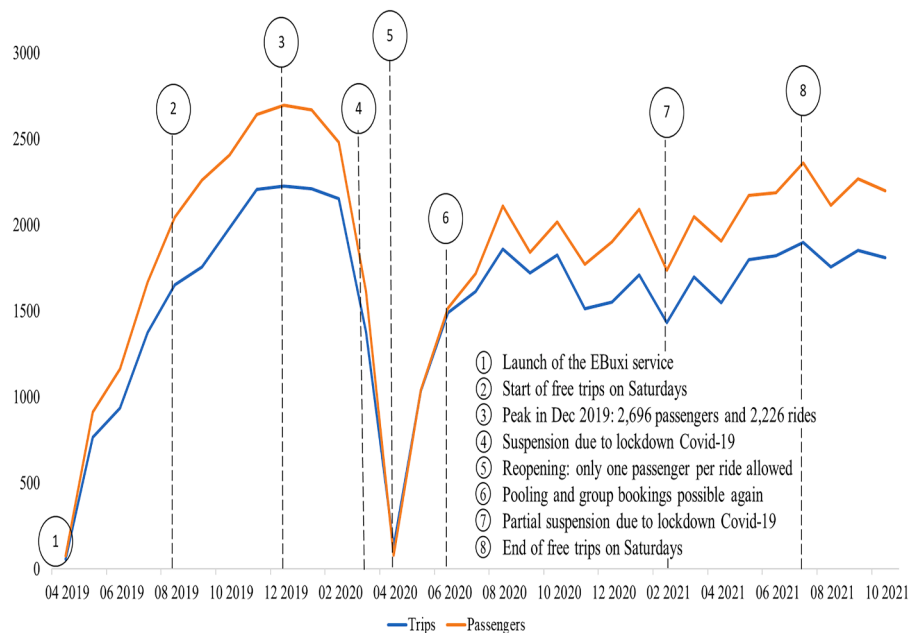


Fig. 2. Demand for EBuxi service.

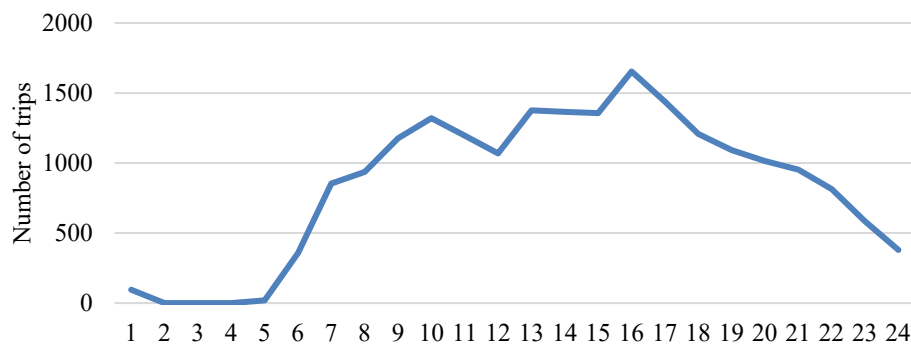


Fig. 3. Time-of-day distribution of trips.

were more tolerant to a longer detour time, followed by the middle-aged and then the younger age group. Only 6.7% could stand more than 15-minute detour. These findings indicate that the users are more tolerant of waiting times than detour times.

5.7. Potential impacts of EBuxi on modal shifts

To evaluate the impact of EBuxi on PT, we asked the user respondents “If the EBuxi service were not available, which alternative modes of transportation would you use for your last EBuxi trip?” (multiple responses possible). Table 6 shows that, without EBuxi, people still made their trips. EBuxi trips mostly replaced walking and cycling trips. The replacement impact of EBuxi on walking and cycling is not necessarily perceived as an undesired consequence. It is instead an indicator that EBuxi serves as a solution to the first-/last-mile problem.

EBuxi trips also replaced a considerable proportion of trips by private cars, either as a driver or as a passenger. For the latter, EBuxi users asked help mainly from family members, relatives, friends or neighbors to give them a ride.

Table 6 indicates that EBuxi does not compete with PT. Only 6.6% of the EBuxi users would take a bus ride if EBuxi was not available. To assess further the impacts of EBuxi on PT, we asked the user respondents

“What means of transportation did you use right before and after your last EBuxi ride for the same journey?” (multiple answers possible). Fig. 6 shows that a majority of respondents (63.5%) used EBuxi as the first travel mode (the first mile) or the last travel mode (the last mile). They mostly used the service to get to or leave the train station. Furthermore, approximately 21% of the users made their journeys only by EBuxi. It is necessary to mention that 17% of the EBuxi user respondents used the service several times per month, almost 12% once a month and the rest occasionally. These results suggest that EBuxi not only complements PT but also helps to increase public transit ridership.

6. Discussion and conclusions

This explorative study has attempted to shed a light on how a specific set of *peri*-urban conditions influences the design and the operation of DRT schemes and their resulting demand patterns and modal shifts. The findings provide insights for planners in developing future DRT systems in PUAs. The key motivation for introducing the EBuxi was to improve accessibility of the non-car population. The provision of PT infrastructure offers a good connection from the PUA’s center to urban towns and cities nearby, but poor PT services in residential areas and between the center and important points of interests. In such *peri*-urban conditions,

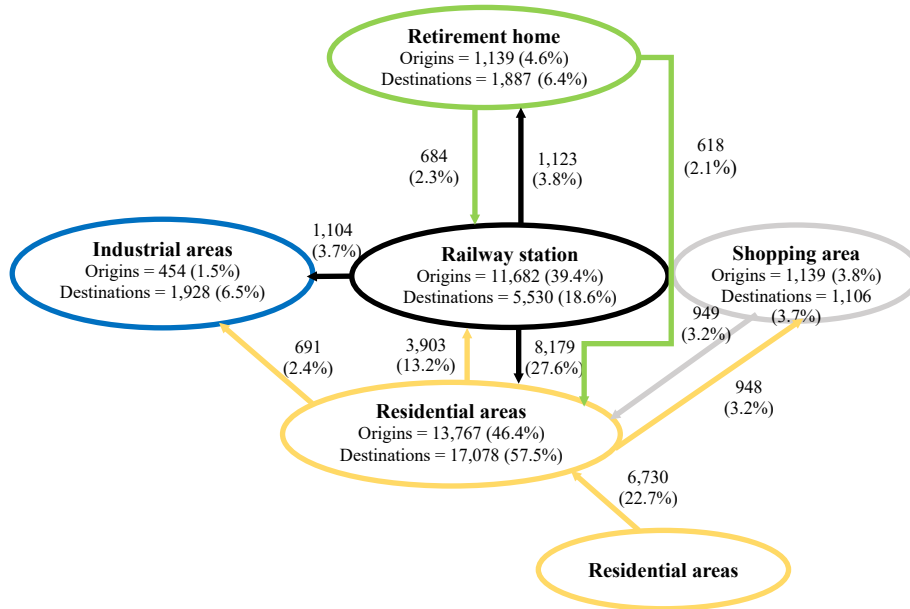


Fig. 4. Distributions of origin–destination flows.

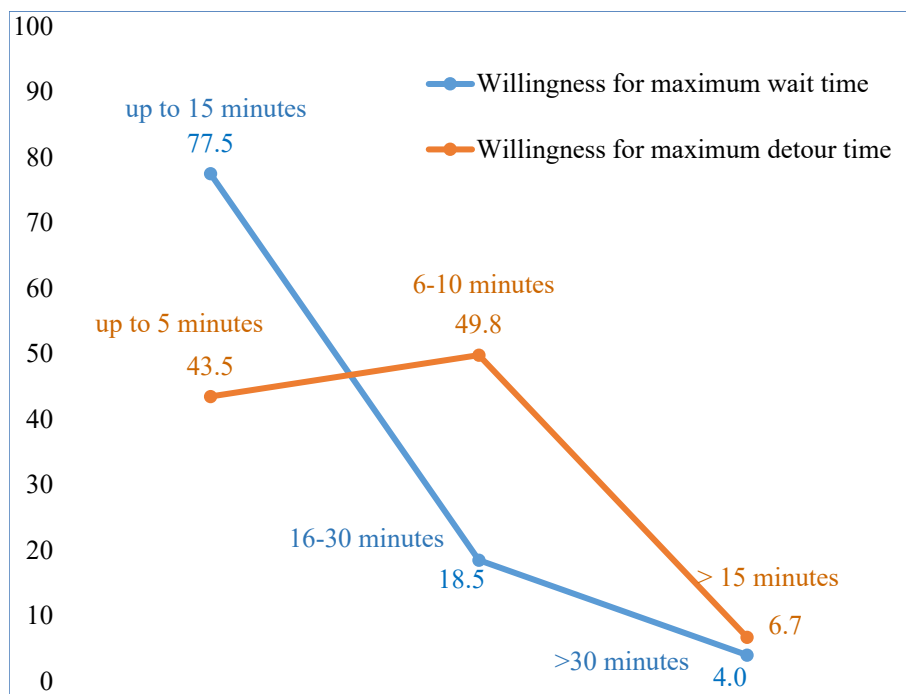


Fig. 5. Willingness to wait and willingness to detour.

Table 6
EBuxi mode replacement impacts.

Replacement travel mode	Responses	Percent
Walk	230	84.9
(E)bike	58	21.8
Car as a passenger	46	17.0
Car as a driver	43	15.9
Bus	18	6.6
Would not have made the trip	8	3.0
n	271	100

an integrated DRT and PT system could improve the accessibility and generate a stable demand while reducing the risk of competing between the two modes of transport. EBuxi was designed as a feeder service to trains and buses. It however offers a great degree of flexibility of routes and schedules. Our empirical evidence shows that EBuxi can help to increase public transit ridership and replace a considerable number of private car trips. For similar reasons, several authors also advocate the integrated DRT and PT system (Alonso-González et al., 2018; Park et al., 2021).

Two lessons can be learned from the EBuxi case study for overcoming the financial risks of DRT schemes. Collaboration among public, private and voluntary stakeholders reduces production, operational and

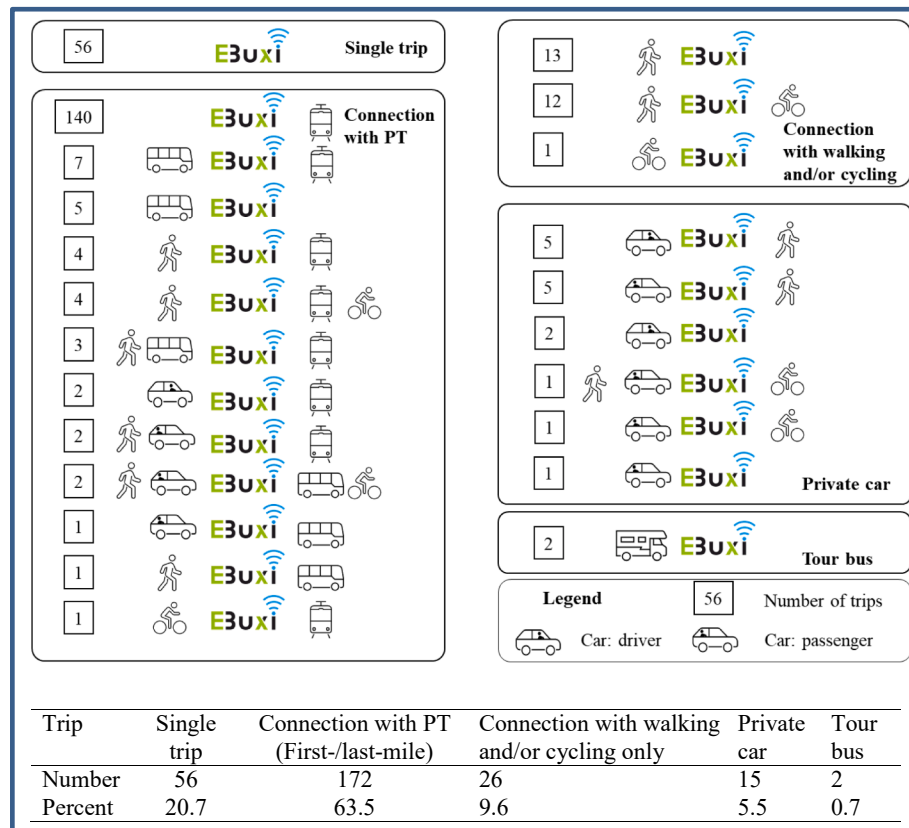


Fig. 6. EBuxi trips in intermodal transportation journeys.

personnel costs, thereby making the fare price more competitive with taxi fares. The engagement of voluntary drivers and the EBuxi association play a central role in this collaboration. The stakeholders could work together effectively when they share a common goal of providing a sustainable transport solution for their own communities and clearly defined responsibilities. The present study supports Docherty et al.'s (2018) claim that a bottom-up solution to DRT services like EBuxi has the potential for success.

The development of demand patterns implies that the DRT service is a niche market, and the break-up of the car dependency is challenging. Despite the increase in the fare during the first two years of operation, the fare change appears not to have affected the demand. The reason for this can be explained as follows. Although the fare was increased, it was still slightly more expensive than a bus fare and considerably cheaper than a taxi fare. Haglund et al. (2019) observe a similar relationship between fare change and demand. A case study of a ride-sharing scheme conducted in rural areas of Switzerland also found that this strategy of fare-pricing contributes to the success of the ride-sharing scheme (Thao et al., 2021). Moreover, the introduction of free-of-charge EBuxi on Saturdays did not boost demand and reduced regular customers' satisfaction with the service and voluntary drivers' motivations. These findings strengthen Thao et al.'s (2021) argument that a free-of-charge mobility on-demand service reduces the appreciation of and contractual binding to the service. The insights regarding fare-pricing are especially valuable to DRT operators that plan to offer flexible, shared DRT services in low-density areas where PT networks are less robust. Furthermore, when a general demand is low and a difference between the off-peak and the peak demand is relatively small, using minibuses could contribute to a cost-effective DRT scheme.

The most frequently reported reasons for not using EBuxi are a preference for cycling or walking or using private cars. This finding indicates that there is no need to turn non-users into users of such shared

on-demand *peri-urban* mobility services as EBuxi. On the contrary, the EBuxi service model could lead to unwanted substitution of walking and cycling. This danger has also been observed in other studies (Haglund et al., 2019; Shamshiripour et al., 2020).

Another contribution of our study is its insights into riders' acceptance of wait times and detour times. To date, only a few studies (e.g. König and Gripenkoven, 2020; Young et al., 2020) have addressed riders' willingness to accept detour times with shared mobility services. DRT users tend to be more tolerant of wait times than detour times. We suggest that using pooling techniques to increase the occupancy rates of DRT services should take into consideration users' acceptance not only of wait times but also of detour times.

In conclusions, the design and the operation of DRT services in the PUA should be tailor-made to its specific conditions of the provision of PT services as well as the interdependencies between service demand, ticket price, fleet utilization and acceptance of wait time and detour time.

Our study did not take into account seasonal effects. This limitation could affect our results as the demand for the DRT service could be higher in winter season than other seasons because walking and cycling is less comfortable.

CRedit authorship contribution statement

Vu Thi Thao: Conceptualization, Formal analysis, Methodology, Visualization, Writing – original draft, Writing – review & editing. **Sebastian Imhof:** Investigation, Formal analysis, Methodology, Visualization, Writing – original draft. **Widar von Arx:** Conceptualization, Investigation, Methodology.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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